



## Selectivity setup in the North Sea sole SS3 model

ICES WGNSSK 2026

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# Problems with current setup

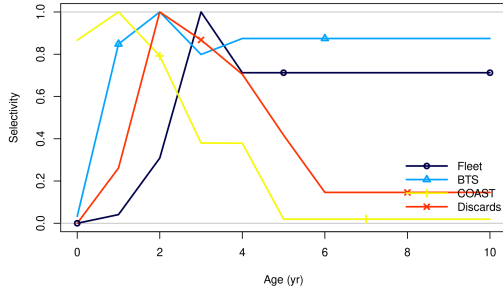
- Current flexibility uses random walks for all fleets, landings changing on time
- Set to a constant value from a given age
- Landings selectivity freely estimated in time
- Final years commonly showing drops in selectivity at age 3-4
- Number of parameters too high

# Changes to selectivity setup

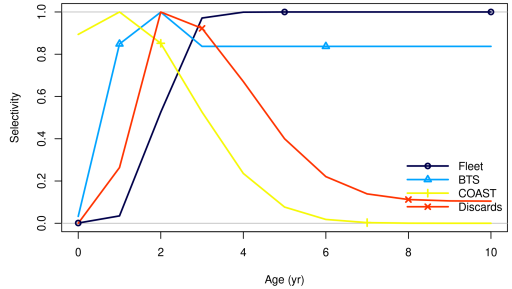
- Random walk per year, fixed from age 5
- Moved to functional forms for all fleets
- Fix a few poorly-estimated parameters
- **Landings**, Logistic
- **BTS** Double normal
  - p4 (descending width), p6 (final), avoid drop at older ages
  - Allowed to vary on pulse years, 2009-2021
- **COAST** Double normal
  - p1 (peak) fixed at  $\sim 0$
- **Discards** Double normal

# Estimated selectivities

- WGNSSK

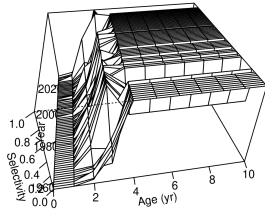


- SELEX

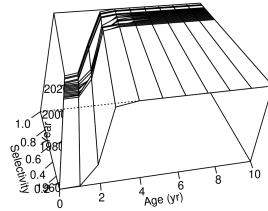


# Landings selectivity

- WGNSSK

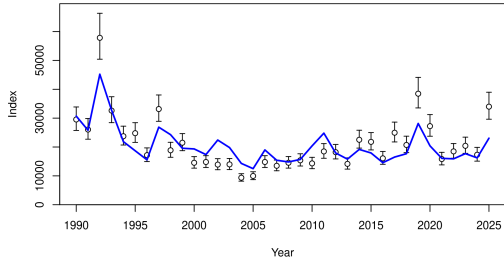


- SELEX

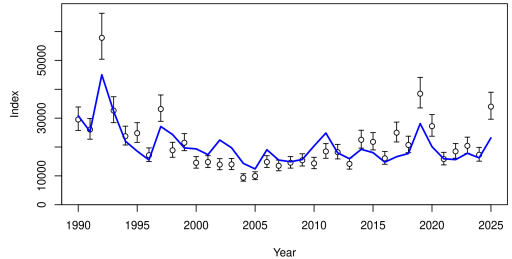


# Fit to BTS

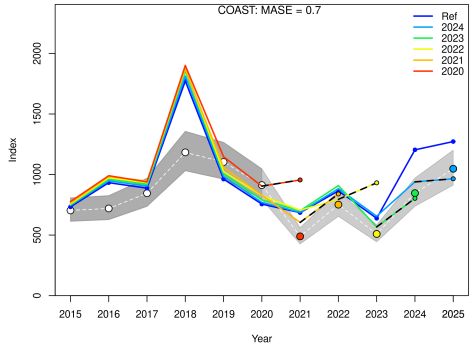
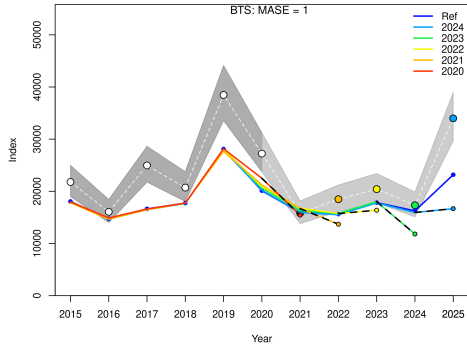
- WGNSSK



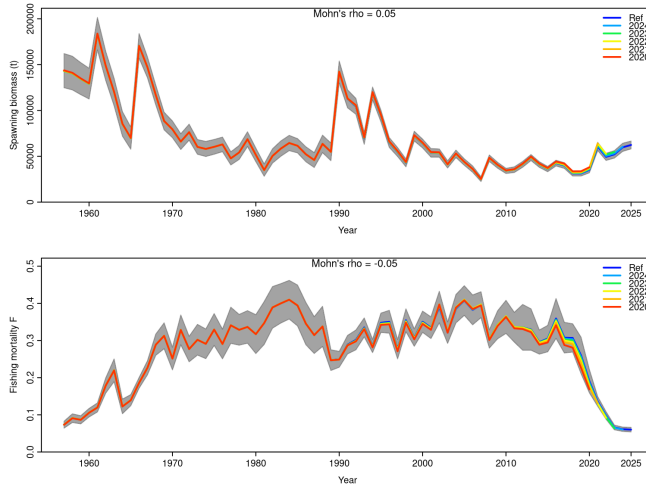
- SELEX



# Prediction skill



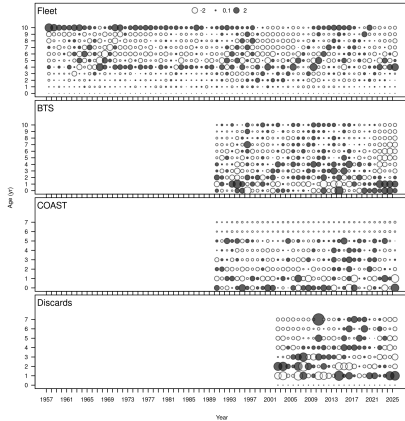
# Retrospective patterns



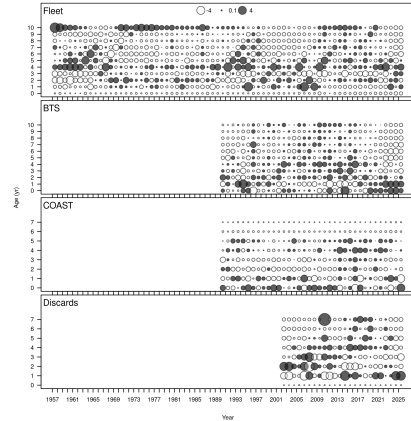


# Residuals

- WGNSSK



- SELEX



# Setup

| #   | LO  | HI       | INIT     | PRIOR    | PR_SD | PR_type | PHASE | env-var | use_dev | dev_mnyr | dev_mxyr | dev_PH | Block | Blk_Fxn | parm_name             |                                  |
|-----|-----|----------|----------|----------|-------|---------|-------|---------|---------|----------|----------|--------|-------|---------|-----------------------|----------------------------------|
| # 1 | 20  | 30       | Fleet    | AgeSelex | 10    | 4       | 0.5   | 5       | 0       | 23       | 2009     | 2021   | 5     | 0       | 0                     | Age_inflection_Fleet(1)          |
|     | -4  | 30       | 0.710311 | 1        | 4     | 0.5     | 6     | 0       | 23      | 2009     | 2021     | 6      | 0     | 0       | Age_95%width_Fleet(1) |                                  |
| # 2 | 9.9 | 1.2036   | BTS      | AgeSelex | 2     | 1       | 0.05  | 4       | 0       | 0        | 0        | 0      | 0.5   | 0       | 0                     | Age_DblN_peak_BTS(2)             |
|     | -5  | 3        | -3.79741 | -5       | 1     | 0.05    | 4     | 0       | 0       | 0        | 0        | 0      | 0.5   | 0       | 0                     | Age_DblN_top_logit_BTS(2)        |
|     | -4  | 12       | -1.44672 | 8.4      | 1     | 0.05    | 4     | 0       | 0       | 0        | 0        | 0      | 0.5   | 0       | 0                     | Age_DblN_ascend_se_BTS(2)        |
|     | -10 | 6        | -3.47201 | 3        | 1     | 0.05    | -4    | 0       | 0       | 0        | 0        | 0      | 0.5   | 0       | 0                     | Age_DblN_descend_se_BTS(2)       |
|     | -15 | 5        | -3.38987 | 0        | 1     | 0.05    | -4    | 0       | 0       | 0        | 0        | 0      | 0.5   | 0       | 0                     | Age_DblN_start_logit_BTS(2)      |
|     | -5  | 15       | 1.63717  | 0.9      | 1     | 0.05    | -4    | 0       | 0       | 0        | 0        | 0      | 0.5   | 0       | 0                     | Age_DblN_end_logit_BTS(2)        |
| # 3 | 9.9 | 4.32E-05 | COAST    | AgeSelex | 2     | 1       | 0.05  | -4      | 0       | 0        | 0        | 0      | 0.5   | 0       | 0                     | Age_DblN_peak_COAST(3)           |
|     | -15 | 3        | -12.5387 | -5       | 1     | 0.05    | 4     | 0       | 0       | 0        | 0        | 0      | 0.5   | 0       | 0                     | Age_DblN_top_logit_COAST(3)      |
|     | -30 | 12       | -20.9735 | 5.9      | 1     | 0.05    | 4     | 0       | 0       | 0        | 0        | 0      | 0.5   | 0       | 0                     | Age_DblN_ascend_se_COAST(3)      |
|     | -2  | 6        | 1.82926  | 0.3      | 1     | 0.05    | 4     | 0       | 0       | 0        | 0        | 0      | 0.5   | 0       | 0                     | Age_DblN_descend_se_COAST(3)     |
|     | -15 | 5        | 1.31219  | 0.8      | 1     | 0.05    | 4     | 0       | 0       | 0        | 0        | 0      | 0.5   | 0       | 0                     | Age_DblN_start_logit_COAST(3)    |
|     | -15 | 5        | -11.8416 | 0        | 1     | 0.05    | 4     | 0       | 0       | 0        | 0        | 0      | 0.5   | 0       | 0                     | Age_DblN_end_logit_COAST(3)      |
| # 4 | 9.9 | 1.19921  | Discards | AgeSelex | 2     | 1       | 0.05  | 4       | 0       | 0        | 0        | 0      | 0.5   | 0       | 0                     | Age_DblN_peak_Discards(4)        |
|     | -15 | 3        | -10.277  | -5       | 1     | 0.05    | 4     | 0       | 0       | 0        | 0        | 0      | 0.5   | 0       | 0                     | Age_DblN_top_logit_Discards(4)   |
|     | -4  | 12       | -3.59095 | 0.3      | 1     | 0.05    | 4     | 0       | 0       | 0        | 0        | 0      | 0.5   | 0       | 0                     | Age_DblN_ascend_se_Discards(4)   |
|     | -2  | 6        | 1.956    | 1        | 1     | 0.05    | 4     | 0       | 0       | 0        | 0        | 0      | 0.5   | 0       | 0                     | Age_DblN_descend_se_Discards(4)  |
|     | -15 | 5        | -14.6525 | 0        | 1     | 0.05    | 4     | 0       | 0       | 0        | 0        | 0      | 0.5   | 0       | 0                     | Age_DblN_start_logit_Discards(4) |
|     | -5  | 5        | -2.14541 | 0.2      | 1     | 0.05    | 4     | 0       | 0       | 0        | 0        | 0      | 0.5   | 0       | 0                     | Age_DblN_end_logit_Discards(4)   |

# Estimated parameters

- WGNSSK

|    | Type                             | Count |
|----|----------------------------------|-------|
| 1  | Total                            | 330   |
| 7  | Stock-recruit                    | 1     |
| 9  | Rec. dev. time series            | 69    |
| 10 | Rec. dev. initial age            | 8     |
| 18 | Age selectivity                  | 18    |
| 19 | Age selectivity time-variation   | 207   |
| 20 | F parameters time series         | 24    |
| 21 | F parameters initial equilibrium | 1     |
| 22 | Dirichlet-Multinomial parameters | 2     |

- SELEX

|    | Type                             | Count |
|----|----------------------------------|-------|
| 1  | Total                            | 148   |
| 7  | Stock-recruit                    | 1     |
| 9  | Rec. dev. time series            | 69    |
| 10 | Rec. dev. initial age            | 8     |
| 18 | Age selectivity                  | 17    |
| 19 | Age selectivity time-variation   | 26    |
| 20 | F parameters time series         | 24    |
| 21 | F parameters initial equilibrium | 1     |
| 22 | Dirichlet-Multinomial parameters | 2     |

# Summary

- Proposed less flexible selectivity setup
- Fewer parameters, more stable estimates
- Fit not dramatically altered
- Are recent fishery changes in need of another block?
- What is happening with BTS?

[https://github.com/iagomosqueira/2026\\_sol.27.4\\_review/selectivity](https://github.com/iagomosqueira/2026_sol.27.4_review/selectivity)

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